

BSR Transition Risk — Methodology

Reflects the code as built through P0–P3. This document specifies the flow, mathematics, and logic of the standalone transition-risk engine (`engine/transition/`) and app (`transition_app/`). It is generated to be marked up: each section states what the code does, the exact formula, the data behind it, and the judgment parameters you can challenge.

Companion to the physical-risk methodology. Screening-grade throughout — not a regulatory disclosure without specialist review.

1. Overview

The engine decomposes the financial impact of decarbonisation on an asset or portfolio into **four channels**, each mapped to one **TCFD transition-risk category**, quantified by one **engine layer**, and routed to exactly one **financial destination** (the *non-duplication* rule):

TCFD category	Layer	Mechanism	Routes to
Policy & Legal	L1	(Scope 1+2) × NGFS carbon price × (1 – pass-through)	Cash-flow OpEx
Technology	L2	Wright's-Law cost crossover / demand collapse → logistic impairment	Cash flow + asset value
Market	L3	Sector shock → Leontief inverse → focal input cost	Cash-flow input cost
Reputation	L4	Sautner CCExposure (z-scored) × per-SD elasticities	Cash flow or WACC

On top of the four layers sit a **decision & assurance layer**: Monte-Carlo uncertainty, tornado sensitivity, PCAF financed emissions, Implied Temperature Rise, PACTA-style alignment, peer benchmarking, marginal-abatement-cost economics, a budget optimizer, and an IFRS S2 / ESRS E1 disclosure export.

How to run

```
pip install streamlit pandas numpy plotly openpyxl
streamlit run transition_app/Home.py           # the app
python -m pytest tests/test_transition.py -q    # the tests (all green)
```

The engine + tests need only numpy/pandas/openpyxl/pytest. `pyam` and `pymrio` are only for the data-refresh scripts (NGFS prices, EXIOBASE MRIO).

2. Architecture & module map

engine/transition/

data_loader.py	cached JSON loaders; ISO3→NGFS region; scenario→NGFS mapping
carbon_pricing.py	L1 – carbon cost, pass-through, abatement pathway, priced fraction
learning_curves.py	L2 – Wright's law, Lafond bands, crossover, logistic impairment
network_propagation.py	L3 – world 20×20 Leontief propagation
network_mrrio.py	L3 – 20×49 EXIOBASE MRIO + Reisch endogenous-default cascade
cc_exposure.py	L4 – z-standardised CCExposure × per-SD elasticities + routing
transition_engine.py	orchestrator: run_asset_transition / run_portfolio_transition
transition_dcf.py	combined physical+transition DCF hook
uncertainty.py	Monte-Carlo over price / pass-through / elasticity
sensitivity.py	one-at-a-time tornado
alignment.py	financed emissions, ITR, pathway alignment, peer benchmark
macc.py	marginal abatement cost curves (decarbonise vs pay)
optimizer.py	budget-constrained merit-order abatement allocation

data/transition/

carbon_prices_ngfs.json	NGFS Phase V REMIND-MAGPIE (real, ×1.18 → USD2020)
sector_pass_through.json	sector-median pass-through
learning_curves.json	learning rates + LCOE base costs (IRENA 2024 / Lazard 2025)
sector_pathways.json	demand pathways (production index, 2025=1.0)
cc_exposure_proxy.json	sector CCExposure (real ×10 ³ scale) + pooled z-base + elasticities
io_matrix.json	20×20 world direct-requirements matrix (EXIOBASE world totals)
io_matrix_mrrio.npz + meta	20×49 EXIOBASE MRIO (980×980)
sector_taxonomy.json	20 sectors: tech pairs, fossil flag, emission intensity
macc.json	per-sector abatement measures (AR6 WG3 / IEA-informed)

Execution flow

- ① Data Entry → intake per entity: sector, Scope 1/2/3, replacement value, revenue, target_year, priced_pct, attribution_pct

|

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run_portfolio_transition(assets, scenarios, ...settings...)

| for each (scenario, asset):

| L1 carbon_cost_timeline(...) → net_carbon_opex[y]

| L2 compute_stranding(...) → impairment[y], revenue_erosion[y]

| L3 propagate (world | mrrio+cascade) → indirect_input_cost[y]

| L4 compute_exposure_premium(...) → revenue_modifier[y] OR wacc_bps

| total_cf[y] = L1 + L2_rev + L3 + L4 ; impairment separate

▼

Results / category pages / Alignment / Abatement / Audit

→ PV, Monte-Carlo P5–P95, tornado, ITR, PCAF, MACC, XLSX + IFRS S2 export

3. Layer 1 — Policy & Legal (direct carbon cost)

Module: `carbon_pricing.py`. For each (scenario, region, year, sector):

3.1 Carbon price

`get_carbon_price(scenario, year, band)` linearly interpolates the NGFS trajectory and holds it flat outside 2025–2050, then multiplies by a Monte-Carlo perturbation `price_scale` (1.0 = base):

```
price(y) = interp(NGFS[scenario][band], y) × price_scale
```

Region **band** ∈ {advanced, emerging, rest_of_world} from `get_ngfs_region(IS03)`. IPCC SSP scenarios map to their closest NGFS analog.

3.2 Abatement pathway (P0)

An entity with a Scope 1+2 net-zero **target_year** decarbonises linearly; carbon cost then applies only to the declining residual:

```
abatement_mult(y) = 1.0                                if no target or y ≤ 2025
                   = residual                            if y ≥ target_year          (residual = re
                   = 1 + (y-2025)/(target-2025)·(residual-1) otherwise
```

3.3 Cost decomposition

```
direct          = (Scope1 + Scope2) × abatement_mult(y)
pass_through    = min(1, sector_PT × pt_scale)           (or per-asset override)
opportunity      = direct × price                        full carbon opportunity cost
passed_through  = opportunity × pass_through             ← L3 market shock (undiminished by free al
net_compliance  = opportunity × priced_fraction         allowances the firm actually buys
absorbed        = net_compliance - passed_through       firm's net margin impact (windfall if < 0)
scope3_indirect = Scope3 × price × (1 - pass_through)
net_carbon_opex = absorbed + scope3_indirect            ← what hits the firm's cash flow
```

- **Windfall economics (Round-1 P5).** The marginal carbon price sets the pass-through opportunity cost regardless of free allocation (Sijm 2012), so pass-through — and the L3 shock — are based on the **full opportunity cost**; `priced_fraction` reduces only the firm's own compliance leg. With generous free allocation and high pass-through, `absorbed` goes **negative** (a windfall gain). At the default `priced_fraction = 1` this reduces to `absorbed = opportunity × (1 - PT)` (legacy).
- **priced_fraction** models free allocation / partial ETS coverage (P0). Default 1.0.
- **Scope-3 mode (P0):** `full` counts `scope3_indirect` in L1 *and* propagates upstream in L3 (reproduces the methodology's worked examples). `auto` sets the L1 Scope-3 term to 0 whenever L3 is enabled, so upstream cost is counted once.

Data: `carbon_prices_ngfs.json` — real **NGFS Phase V** (released Nov 2024) REMIND-MAgPIE 3.3-4.8 prices pulled live from the IIASA `ngfs_phase_5` explorer, rebased $\times 1.18$ (US GDP deflator 2010 $\rightarrow$ 2020) to $\sim$ USD2020. Six scenarios carry real data; `divergent_net_zero` + three IEA scenarios retain Appendix-D placeholders. (*Round-1 P4: the Kotz et al. 2024 retraction affects only NGFS physical-damage variables, not the transition/price pathways — L1 uses standard REMIND-MAgPIE transition price variables and is unaffected.*) **Pass-through:** `sector_pass_through.json`, sector medians (Sijm 2012; Fabra & Reguant 2014; Cludius 2020).

4. Layer 2 — Technology (learning curves & stranding)

Module: `learning_curves.py`.

4.1 Wright's Law

$$\begin{aligned} \text{Cost}(Q) &= \text{Cost}_0 \times (Q/Q_0)^b, & b &= \log_2(1 - \text{LR}) \\ Q/Q_0 &= (1 + g)^{(y - 2025)}, & g &= \max(0, \text{scenario deployment growth}) \end{aligned}$$

Negative growth (demand decline) is clamped to 0 — cumulative capacity is monotonic; demand decline is handled by the sector pathway, not the cost curve.

4.2 Lafond distributional band

$$\begin{aligned} \sigma_t &= \text{lafond}_\sigma \times \sqrt{(y - 2025)} \\ \text{cost_lo} &= \text{pred} \times e^{(-\sigma_t)}, & \text{cost_hi} &= \text{pred} \times e^{(+\sigma_t)} \end{aligned}$$

4.3 Stranding trigger (either fires)

- **Cost crossover** — first year challenger cost $\leq$ incumbent cost.
- **Demand collapse** (fossil-dependent sectors only) — first year the sector pathway falls below 0.5.

4.4 Impairment

$$\begin{aligned} \text{frac}(y) &= 1 / (1 + e^{(-0.20 \cdot (y - \text{crossover}))}) && \text{logistic, slope } 0.20 \\ \text{base} &= \text{replacement_value} && \text{if fossil-dependent} \\ &= \text{replacement_value} \times 0.5 && \text{otherwise (only incumbent-tied value can strand)} \\ \text{cap} &= \max(0, 1 - \text{pathway}(2050)) \times \text{base} && \text{scenario-scaled ceiling} \\ \text{annual_impairment}(y) &= (\text{frac}(y) - \text{frac}(y-1)) \times \text{cap} \end{aligned}$$

Impairment is a **balance-sheet** figure, reported separately from cash flow.

4.5 Revenue erosion

```
revenue_index(y) = sector_pathway(y)          (production index, 2025 = 1.0)
revenue_erosion(y) = max(0, revenue × (1 - index(y))) ← cash-flow cost
```

Data (P0 refresh): power-sector base costs are **LCOE (USD/MWh)** from **IRENA Renewable Power Generation Costs in 2024** and **Lazard LCOE+ 2025 v18** (coal 118, gas 76, solar 43, onshore wind 34, storage 170). Learning rates: Way et al. 2022 / IRENA. Industrial base costs remain sector estimates.

5. Layer 3 — Market (production-network propagation)

Modules: `network_propagation.py` (world), `network_mrio.py` (MRIO + cascade).

5.1 Sectoral shock

Each sector's carbon shock as a fraction of its output:

```
si = emission_intensityi × price × pass_throughi × 1e-6
```

(emission_intensity in tCO₂ per M\$ revenue; 1e-6 → per-USD.) Using sector-typical exposure avoids double-counting the focal asset's own pass-through.

5.2 Leontief propagation

```
L = (I - A)-1
total_shock_j = Σi L[i,j] · si
propagated_j = total_shock_j - own      (own = L[j,j]·sj, already in L1)
indirect_cost = propagated_j × asset_revenue
```

CES damping: off-diagonal A scaled by 1/σ before inversion (σ = substitution elasticity; Papageorgiou 2017 range 1.3–3.0). σ=1 is Cobb-Douglas.

5.3 Resolution (P2)

- **World** (default): 20×20 single-region matrix (EXIOBASE-3 world totals, `io_matrix.json`).
- **Multi-region:** full **20 sectors × 49 regions = 980×980** EXIOBASE MRIO (`io_matrix_mrio.npz`). Cross-border supply chains are explicit and each region's sectors pay carbon on **that region's price band**, so an EU refiner's indirect cost reflects emerging-market upstream at a different carbon price. Leontief inverts in ~0.1 s.

5.4 Endogenous-default cascade (P2, Reisch et al. 2025)

Beyond linear Leontief, nodes absorbing input-cost shock above threshold **θ** pass an amplified shock downstream, iterated to convergence:

```
s_eff = s
repeat: total = L · s_eff ; over = max(0, total - θ) ; s_eff = s + contagion · over
```

Default off; triggers only for the most-exposed nodes under stress ($\theta \approx 0.02 \rightarrow \sim 1.25\times$ for CHN steel; $\theta = 0.01 \rightarrow \sim 6.8\times$).

6. Layer 4 — Reputation (CCExposure, z-standardised)

Module: `cc_exposure.py`. **P0 fix (D3):** raw 0–2 proxies inflated elasticities $\sim 500\text{--}1000\times$; the engine now consumes **z-scores** against the real Sautner distribution and prices **per standard deviation**.

```
raw_x = sector_exposure_x × scenario_modifier_x          x ∈ {opp, reg, phy}
z_x   = (raw_x - pooled_mean_x) / pooled_sd_x
z_total = (Σ raw_x - pooled_mean_total) / pooled_sd_total

credit_spread_bps = 12 · z_reg + 6 · z_phy
equity_premium_bps = 50 · z_total                      (Pricing paper: premium per 1 SD of overall)
revenue_growth_bps = 35 · z_opp - 25 · z_reg
```

The **pooled distribution** is Sautner et al. (JoF 2023) **Table 1** (firm-year, $\times 10^3$): opp 0.391/1.344, reg 0.049/0.264, phy 0.013/0.103, total 0.943/2.443. **Sector exposures** are anchored to Sautner **Table 4** by SIC industry where available (utilities $\rightarrow$ power, petroleum refining, transport equipment $\rightarrow$ autos, primary metal $\rightarrow$ steel ...), estimated from adjacent industries otherwise.

Effect: average-exposure sectors carry ≈ 0 premium (the inflation is gone); coal keeps ~ 29 bps credit / -39 bps revenue drag; renewables $+85$ bps revenue uplift.

Routing (non-duplication): `ROUTE_CASHFLOWS` (default) applies the revenue-growth modifier to cash flows (credit/equity are diagnostic); `ROUTE_WACC` adds credit+equity to the discount rate and zeroes the revenue modifier. **Firm override (P2):** a licensed firm-level CCExposure feed replaces the sector median via `firm_override`.

7. Orchestration & non-duplication

`run_asset_transition(asset, scenario, ...)` composes the layers. Each channel routes once:

```
total_cf(y) = L1 net_carbon_opex
              + L2 revenue_erosion
              + L3 indirect_input_cost
              + L4 revenue_modifier (as a cost: -modifier)
impairment(y) = L2 annual_impairment          (balance-sheet, NOT in total_cf)
wacc_premium_bps = L4 credit+equity          (only if ROUTE_WACC)
```

`run_portfolio_transition` runs all assets × scenarios and returns `{scenario: [TransitionAssetResult]}`.

Present value & discount basis (Round-1 P2). Carbon prices are **real** (USD2020), so PV discounts the real cash flows at a **real** rate = nominal WACC – long-run inflation ($PV = \sum \text{cost} / (1 + \text{real})^{(y - 2025)}$). Discounting real flows at the nominal WACC would systematically understate PV. Both are app inputs (nominal WACC default 9%, inflation 2.5% → real 6.5%).

Impairment vs cash-flow cost are NOT additive (Round-1 P1). L2 stranded impairment is the PV writedown of the same future cash flows whose erosion already feeds the cash-flow cost. They are two **lenses on one loss** — reported side by side, never summed into a combined total (enforced in the dashboard, XLSX, and disclosure export; guarded by a test).

8. Uncertainty & sensitivity

8.1 Monte-Carlo (`uncertainty.py`)

Re-runs the four layers over N draws (default 400), perturbing:

<code>price_scale</code>	~ LogNormal(0, 0.20)	carbon-price / policy path
<code>pass_through_scale</code>	~ Normal(1, 0.10)	cost incidence
<code>elasticity σ</code>	~ Uniform(1.0, 2.5)	network substitution

Returns P5 / P50 / P95 of PV transition cost and stranded impairment. Seed-deterministic.

8.2 Tornado (`sensitivity.py`)

One-at-a-time swing of each driver (carbon price ±30%, pass-through ±20%, σ 1.0–2.5, Scope-3 mode, discount ±2pp) with all others at base; sorted by |swing|. Attributes the Monte-Carlo spread to individual assumptions so you know which input to firm up.

9. Decision & alignment layer

9.1 Financed emissions — PCAF (`alignment.py`)

Attributed emissions = $\sum \text{attribution_pct} \times (\text{Scope } 1/2/3)$. 100% = corporate own-asset view; a lender/investor enters their stake.

9.2 Implied Temperature Rise

Each entity's abated Scope 1+2 pathway vs a 1.5 °C linear-to-net-zero budget:

```

budget = 0.5 × E0 × (2050 – 2025)           area under a straight line to zero
ratio  = cumulative_actual / budget – 1
ITR    = clamp[1.2, 4.0]( 1.5 + 1.2 × ratio )
portfolio ITR = Scope1+2 × attribution weighted mean

```

Net-zero-by-2050 ≈ 1.5 °C; flat emissions ≈ 2.8 °C.

9.3 Pathway alignment (PACTA-style)

Compares the entity's Scope 1+2 decline by 2050 to the scenario's sector demand pathway; gap ≤ 0.05
→ aligned, ≤ 0.25 → lagging, else misaligned.

9.4 Peer benchmarking

Entity economic intensity (tCO₂ Scope 1+2 / \$M revenue) vs the **real Sautner firm-level Carbon Intensity distribution** (JoF 2023 Table 1: median 11, p75 84.6 across ~10k firms) → quartile position.

9.5 Marginal abatement cost curves (`macc.py`)

Per-sector measures (cost USD/tCO₂, potential share of Scope 1+2). At carbon price P, all measures with cost ≤ P are cost-effective:

```

abated_frac = Σ potential (cost ≤ P) , capped at 1
abatement_cost = Σ cost × potential × emissions
carbon_cost_avoided = abated × P , net_benefit = avoided – cost

```

9.6 Budget optimizer (`optimizer.py`)

Merit-order greedy: sort all measures across the portfolio by \$/tCO₂; fund no-regret (negative-cost) measures always, then buy cheapest-first until the budget is exhausted. Returns abated tonnes, net spend, marginal-cost frontier, residual carbon cost, and per-asset allocation. Optimal for max tonnes per dollar under a linear MACC.

9.7 Disclosure export

`build_disclosure_report` generates a Governance → Strategy → Risk Management → Metrics report mapped to **IFRS S2** clauses and **ESRS E1** datapoints, plus a provenance-stamped multi-sheet XLSX.

10. Data provenance & vintages

Component	Source	Vintage
Carbon prices (L1)	NGFS Phase V REMIND-MAGPIE 3.3-4.8 (IIASA)	Nov 2023, ×1.18 → USD2020
Pass-through (L1)	Sijm 2012; Fabra & Reguant 2014; Cludius 2020	sector medians
Learning rates (L2)	Way et al. 2022; IRENA	2022–23
LCOE base costs (L2)	IRENA RPGC 2024; Lazard LCOE+ 2025 v18	2024–25
I-O matrix (L3)	EXIOBASE-3 (Stadler et al. 2018), IOT 2019	2019
CCExposure + z-base (L4)	Sautner et al. JoF 2023 Tables 1 & 4	2002–2019, 10k firms
Sector pathways	NGFS / IEA WEO 2023	2023
MACC	IPCC AR6 WG3; IEA roadmaps	2022–23

11. Calibration — worked examples (default settings, Scope-3 full)

- **Coal plant** (\$500M, 2.5 MtCO₂): Net-Zero-2050 L1-2050 ≈ **\$169M**; cumulative impairment **\$266.1M** / 97.3% stranded (crossover 2025). Current Policies ≈ \$7M / \$33M.
- **Oil refinery** (\$2B, 12.8 MtCO₂): stranding triggered by **demand collapse** (crossover 2039), not cost crossover.
- **Office** (\$50M, 1k tCO₂): no stranding; small net cost / reputational uplift.

These are unchanged by the P0–P3 upgrades (abatement/priced default to legacy; world L3 default; LCOE preserves the 2025 crossover).

12. Known limitations (challenge these)

- **Pass-through** is a static sector median; firm-specific values vary with market power and trade exposure. Override per asset.
- **CCExposure** sector exposures are anchored to published SIC-industry means, estimated for sectors not in Table 4; the licensed firm-level feed is the fix.
- **MRIO twin sectors** (`road_transport_ev` , `real_estate_residential`) are supply-split from their siblings (EXIOBASE has no separate EV / residential-RE industry).
- **Cascade 0 / contagion** are judgment parameters; the linear result is the default.
- **ITR / MACC / peer benchmark** are screening approximations aligned to SBTi/PACTA/PCAF/AR6 in spirit, not certified implementations.
- **Carbon-price vintage** is NGFS Phase V (Nov 2024); refresh for disclosure.
- Emissions decarbonise only if a target is set; without one, Scope 1+2 are held flat.
- **Structural simplifications (Round-1 U3)**: (i) IAM shadow prices are treated as *realised* carbon prices — a first-best assumption that overstates cost where policy underdelivers; (ii) the default world 20×20 matrix carries I-O aggregation bias; (iii) three price bands (advanced/emerging/RoW) proxy NGFS's ~12 model regions (the 20×49 MRIO mode relaxes this).

- **Endogenous-default cascade (Round-1 U2)** rests on Reisch et al. 2025, an un-peer-reviewed preprint — research-grade only; default-off and excluded from the disclosure export.
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Appendix A — Sector taxonomy (20 sectors)

Key · label · incumbent → challenger · fossil-dependent · emission intensity (tCO₂/M\$). See `data/transition/sector_taxonomy.json`. 20 sectors span power (coal/gas/renewable), oil & gas (upstream/refining/distribution), heavy industry (steel/cement/chemicals/aluminium), transport (road ICE/EV, aviation, shipping), real estate (commercial/residential), agriculture, data centres, general manufacturing, and services.

Appendix B — Pass-through coefficients

See `data/transition/sector_pass_through.json` (pass-through, demand elasticity, market structure, trade exposure, source).

Appendix C — Learning rates & LCOE

See `data/transition/learning_curves.json`. Power LCOE (USD/MWh): coal 118, gas 76, solar 43, onshore wind 34, storage 170 (IRENA 2024 / Lazard 2025).

Appendix D — NGFS carbon prices

See `data/transition/carbon_prices_ngfs.json`. Real REMIND-MAgPIE, three bands, 2025–2050.

Appendix E — Layer-4 z-base & elasticities

Pooled distribution (Sautner Table 1, $\times 10^3$): opp 0.391/1.344 · reg 0.049/0.264 · phy 0.013/0.103 · total 0.943/2.443. Per-SD elasticities (bps): equity 50·z_{total}; credit 12·z_{reg} + 6·z_{phy}; revenue 35·z_{opp} – 25·z_{reg}.

Appendix F — MACC measures

See `data/transition/macc.json`. Per-sector measures with marginal cost (USD/tCO₂) and abatement potential (share of Scope 1+2); AR6 WG3 / IEA-informed.

Appendix G — References

NGFS Phase V (IIASA) · Sijm 2012 · Fabra & Reguant 2014 · Cludius 2020 · Way et al. Joule 2022 · Lafond et al. TFSC 2018 · IRENA RPGC 2024 · Lazard LCOE+ 2025 · Stadler et al. (EXIOBASE) 2018 · Acemoglu et al. 2012 · Papageorgiou et al. 2017 · Reisch et al. 2025 (arXiv:2503.10644) · Sautner, van Lent, Vilkov, Zhang (JoF 2023; Mgmt Sci 2023) · IPCC AR6 WG3 · PCAF 2022 · SBTi · PACTA.